

figures-referenced

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
library(dplyr)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))
```

```

gradient_it <- read_csv(here("data", "it_gradient_data.csv"))

metadata <- read_csv(here("data", "metadata.csv"))

site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

trail_creek_isco <- compiled_isotope |>
  filter(site %in% c("utc_isco", "tc_isco"))

trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc"))

italian_creek <- compiled_isotope |>
  filter(location == "it")

trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_std,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_std) |>
  mutate(date = mdy(date)) # telling r the date format

utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

tc_isco <- trail_creek_isco |>
  filter(site == "tc_isco")

# Downstream Trail Creek
tc_metrics <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"), show_col_types = FALSE)
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,

```

```

    temp = temperature_c,
    water_level = water_level_m,
    bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_metrics <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types = FALSE) |> #
clean_names() |> # lowercase and underscores in column names
rename(date_time = date_time_mdt, # renaming columns to be shorter
       dp = differential_pressure_k_pa,
       ap = absolute_pressure_k_pa,
       temp = temperature_c,
       water_level = water_level_m,
       bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "utc") # creating a new column of the site

tc_utc_metrics <- merge(tc_metrics, utc_metrics, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_metrics,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data

```

```

  by.y = c("location", "date") # matching columns in compiled_isotope
)

discharge_tc_isco <- read_csv(here("data", "discharge_tc_isco.csv"))

discharge_utc_isco <- read_csv(here("data", "discharge_utc_isco.csv"))

discharge_isco <- read_csv(here("data", "isco_discharge_lc.csv"))

hydrograph <- read_csv(here("data", "hydrograph.csv")) |>
  mutate(date = mdy(date)) # telling r the date format

```

Figures

Figure 1: Time-Series of Line-Conditioned Excess (lc-excess) of ISCO's

```

ggplot(data = trail_creek_isco,
  aes(x = date,
    y = lc_excess,
    color = site,
    group = site)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 1.5,
    alpha = 0.8) +
  geom_line(linewidth = 1.0) +
  scale_color_manual(
    values = c("tc_isco" = "dodgerblue", "utc_isco" = "orangered2"),
    labels = c("tc_isco" = "Trail Creek", "utc_isco" = "Upper Trail Creek")
  ) +
  labs(title = "Trail Creek ISCO: Line-Conditioned Excess Over Time",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "lc-excess (\u2030)",
    color = "Site") +
  ylim(-6,0) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),

```

```
panel.grid.minor = element_blank(),  
legend.position = "top")
```

Trail Creek ISCO: Line-Conditioned Excess Over Time

Values below 0 indicate potential evaporative enrichment

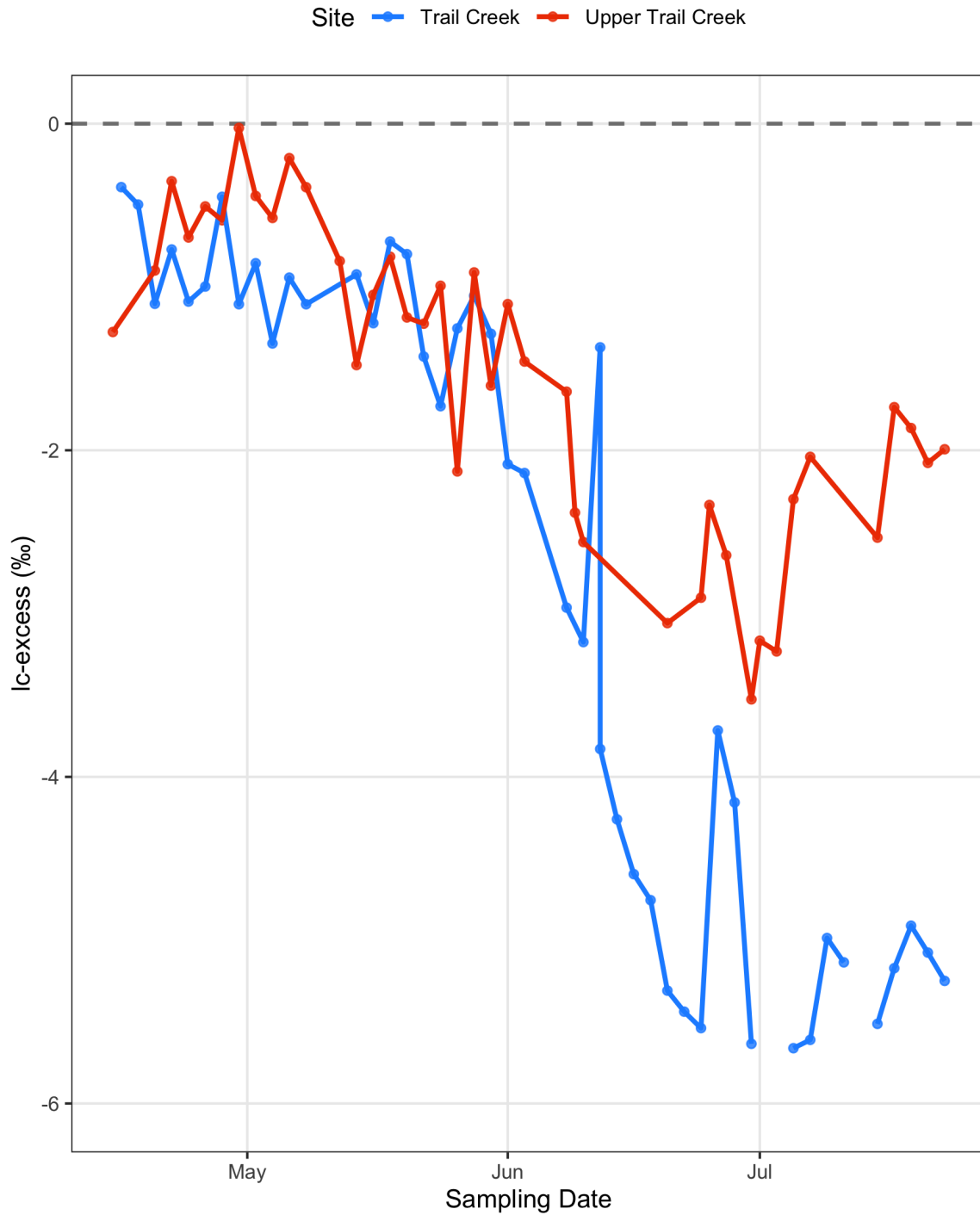


Figure 2: Sampling Site and lc-excess (Trail Creek)

```
# Standardize names to match your raw data (e.g., tc_us, tc_mid...)
distances_tc <- tibble(
  site = c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3"),
  dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)
)

# Add the physical distance column to your data frame
trail_creek_mapped <- trail_creek |>
  left_join(distances_tc, by = "site")

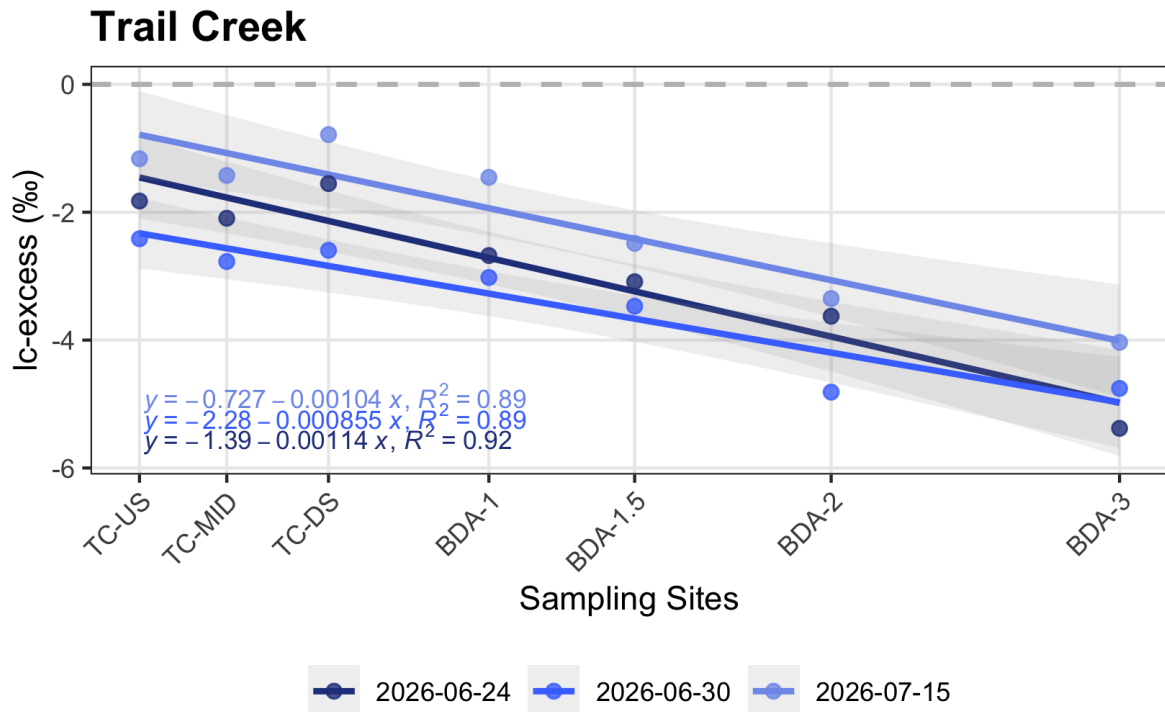
plot2_1 <- ggplot(data = trail_creek_mapped, aes(x = dist_m, y = lc_excess, color = factor(d
# reference line at 0
geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
# linear regression lines using real distances
geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
# Equations will now display correct physical slopes (per meter)
stat_poly_eq(
  aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\\", \\"*")),
  formula = y ~ x,
  parse = TRUE,
  label.x = "left",
  label.y = "bottom",
  vstep = 0.05,
  size = 3.0) +
# Points plotted at exact stream distances
geom_point(size = 2.0, alpha = 0.8) +
labs(
  title = "Trail Creek",
  x = "Sampling Sites",
  y = "lc-excess (\\u2030)",
  color = "Sampling Date"
) +
scale_color_manual(values = c("2026-06-24" = "royalblue4", "2026-06-30" = "royalblue1", "2
# Continuous scale maps site labels directly to their physical meter marks
scale_x_continuous(
  breaks = c(57.6, 333, 654, 1160, 1620, 2240, 3150),
  labels = c("TC-US", "TC-MID", "TC-DS", "BDA-1", "BDA-1.5", "BDA-2", "BDA-3")
) +
theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
```

```

panel.grid.minor = element_blank(),
axis.text.x = element_text(angle = 45, hjust = 1),
legend.position = "bottom",
legend.title = element_blank()
)

```

plot2_1



```

distances_cc_lm <- tibble(
  site = c("coal_1", "coal_2", "coal_4", "coal_5", "coal_6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)
)

# 2. Join the continuous distance column into your dataset
coal_creek_mapped <- coal_creek |>
  left_join(distances_cc_lm, by = "site")

# 3. Plot using continuous physical distance on the x-axis
plot2_2 <- ggplot(data = coal_creek_mapped, aes(x = dist_m, y = lc_excess, color = factor(date))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +

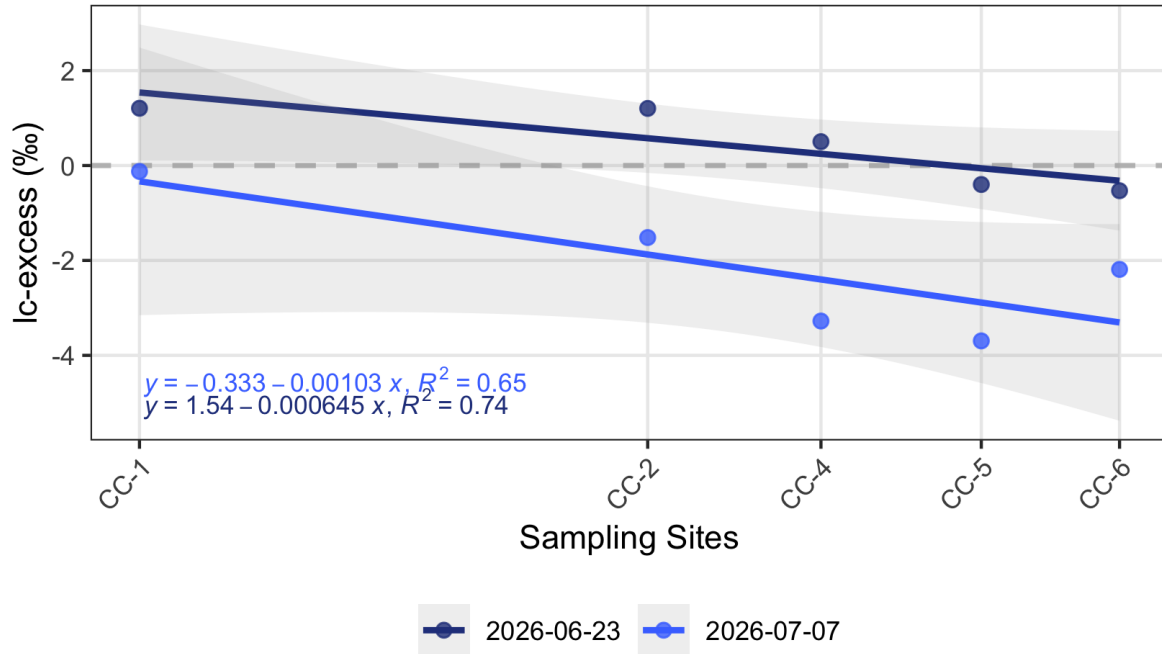
```

```

geom_smooth(method = "lm", formula = y ~ x, se = TRUE, linewidth = 1, alpha = 0.15) +
stat_poly_eq(
  aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\\", \\"*")),
  formula = y ~ x,
  parse = TRUE,
  label.x = "left",
  label.y = "bottom",
  vstep = 0.05,
  size = 3.0
) +
geom_point(size = 2.0, alpha = 0.8) +
scale_color_manual(values = c("2026-06-23" = "royalblue4", "2026-07-07" = "royalblue1")) +
# Map labels to their precise coordinate positions
scale_x_continuous(
  breaks = c(0, 1496, 2006, 2478, 2885),
  labels = c("CC-1", "CC-2", "CC-4", "CC-5", "CC-6")
) +
labs(
  title = "Coal Creek",
  x = "Sampling Sites",
  y = "lc-excess (\\u2030)",
  color = "Sampling Date"
) +
theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom",
  legend.title = element_blank()
)
plot2_2

```

Coal Creek



```
distances_it_lm <- tibble(
  site = c("italian_1", "italian_2", "italian_3", "italian_4", "italian_5"),
  dist_m = c(0, 793.37, 1665.83, 2385.13, 2963.62)
)

italian_creek_mapped <- italian_creek |>
  left_join(distances_it_lm, by = "site")

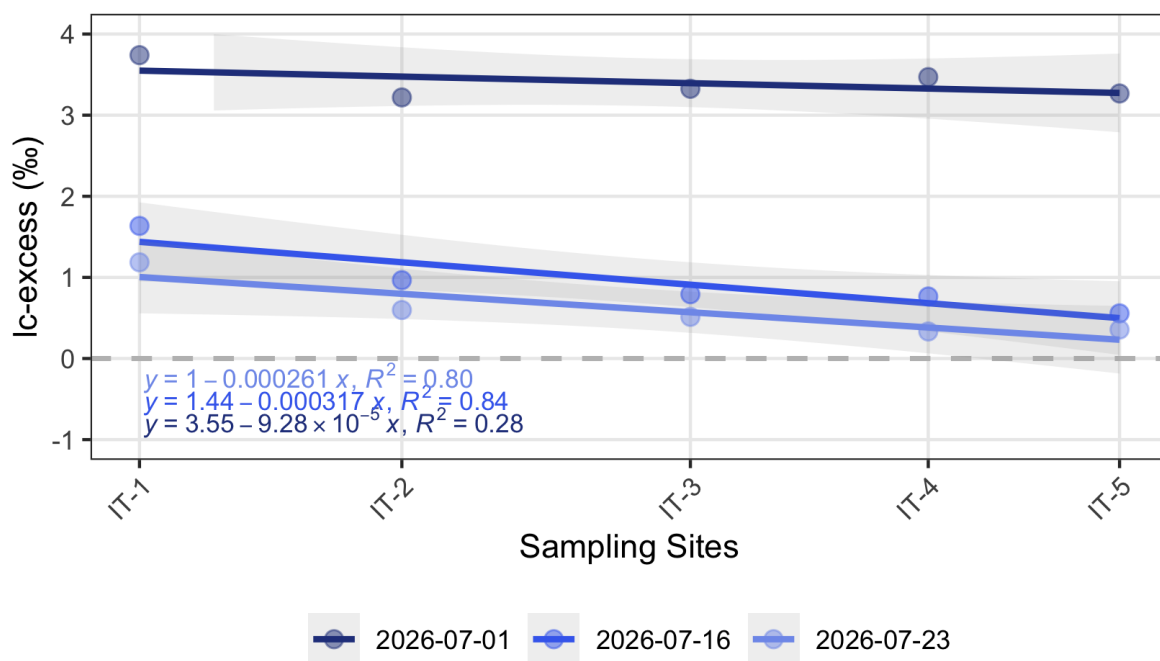
plot2_3 <- ggplot(data = italian_creek_mapped, aes(x = dist_m, y = lc_excess, color = factor(
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, linewidth = 1, alpha = 0.15) +
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\\", \\"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left",
    label.y = "bottom",
    vstep = 0.05,
    size = 3.0
  ) +
```

```

ylim(-1, 4) +
geom_point(size = 2.5, alpha = 0.5) +
scale_color_manual(values = c("2026-07-01" = "royalblue4", "2026-07-16" = "royalblue2", "20
# Map labels to their precise coordinate positions
scale_x_continuous(
  breaks = c(0, 793.37, 1665.83, 2385.13, 2963.62),
  labels = c("IT-1", "IT-2", "IT-3", "IT-4", "IT-5")
) +
labs(
  title = "Italian Creek (control)",
  x = "Sampling Sites",
  y = "lc-excess (\u2030)",
  color = "Sampling Date"
) +
theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom",
  legend.title = element_blank()
)
plot2_3

```

Italian Creek (control)



```
figure_2 <- plot2_3 + plot2_1 + plot2_2
```

```
figure_2
```

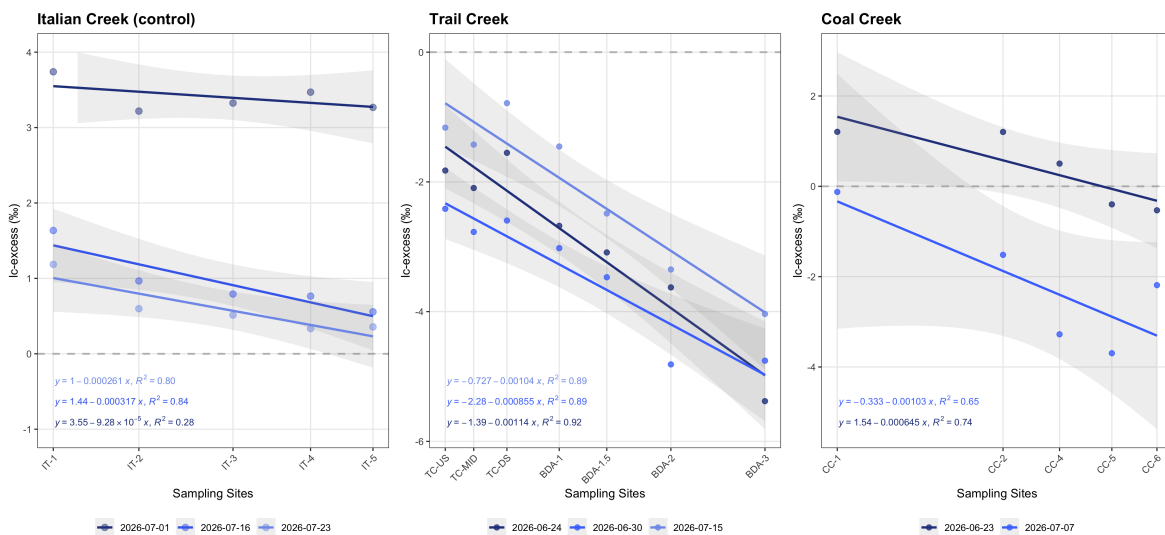


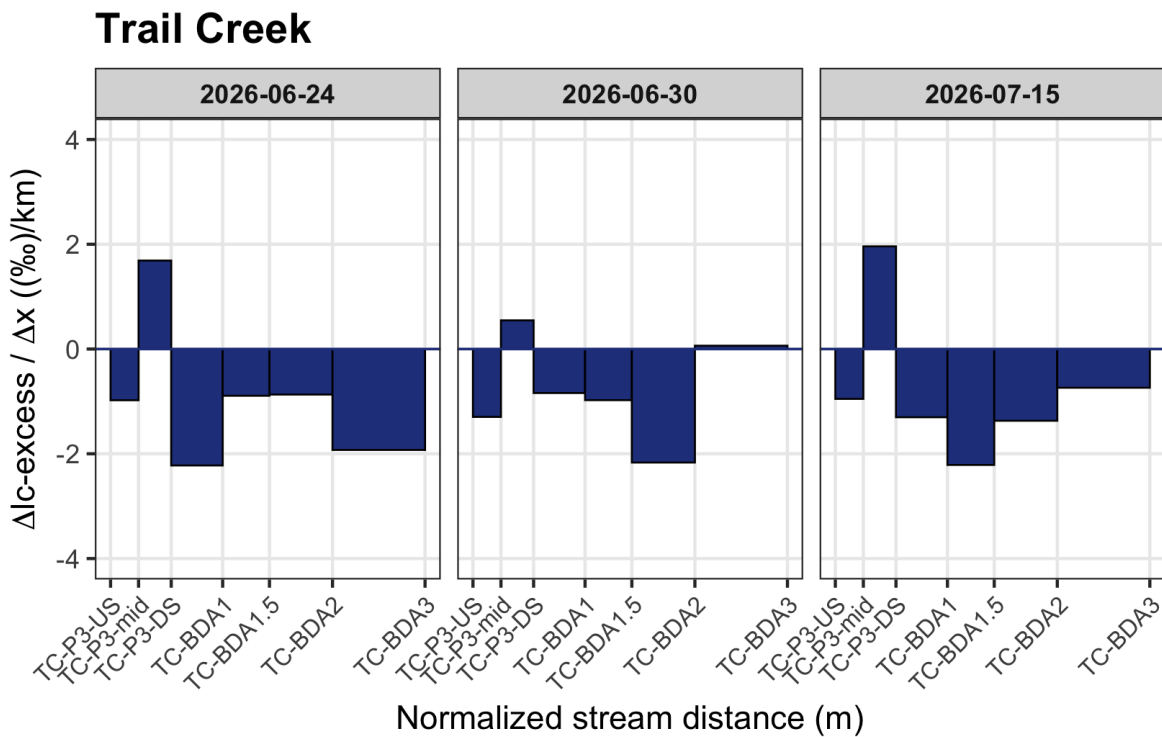
Figure 3:

```
# Spatial variations in  $\Delta$ lc-excess/ $\Delta$ x across sampling dates
# Buildings a look up table of stream distances Trail Creek
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_tc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`),
               names_to = "date",
               values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                           date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                           date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
plot3_1 <- ggplot(gradient_tc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue4", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue4", linewidth = 0.4) +
  ylim(-4,4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
```

```
axis.text.x = element_text(angle = 45, hjust = 1, size = 8))
plot3_1
```



```
# Building a look up table of stream distances Coal Creek
distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
mutate(xmax = dist_m,
       xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
gradient_cc_clean <- gradient_cc |>
select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`) |>
pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
             names_to = "date",
             values_to = "gradient") |>
mutate(date = case_when(
  date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
  date == "Δlc-excess/Δx_7-7" ~ "2026-07-07",
  gradient = as.numeric(gradient)) |>
```

```

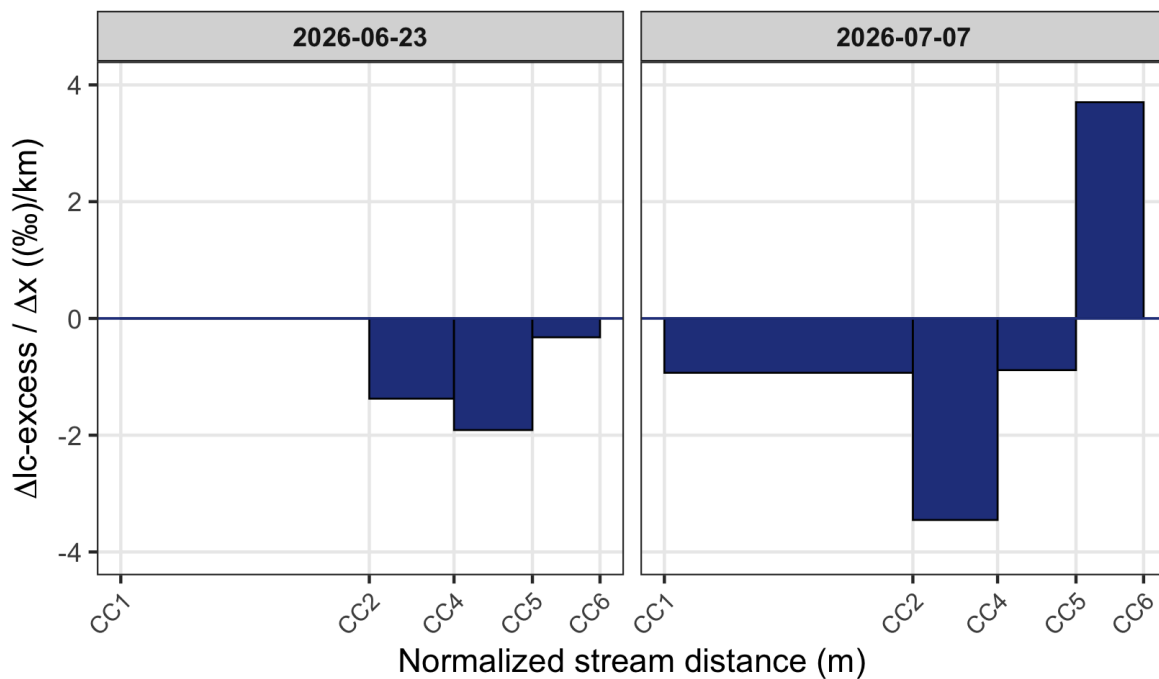
left_join(distances_cc |> select(site, xmin, xmax),
          by = "site") |>
mutate(site = factor(site, levels = distances_cc$site)) |>
filter(!is.na(gradient))

# Gradient plot
plot3_2 <- ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue4", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue4", linewidth = 0.4) +
  ylim(-4,4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8))

plot3_2

```

Coal Creek



```
# Building a look up table of stream distances Coal Creek
```

```
distances_it <- tibble(
  site = c("IT1", "IT2", "IT3", "IT4", "IT5"),
  dist_m = c(0, 793.37, 1665.83, 2385.13, 2963.62)
) |>
mutate(
  xmax = dist_m,
  xmin = lag(dist_m, default = 0)
)
```

```
# Creating a filtered data frame for lc-excess
```

```
gradient_it_clean <- gradient_it |>
select(site, normalized_stream_distance_m, `d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`)
pivot_longer(
  cols = c(`d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`),
  names_to = "date",
  values_to = "gradient"
) |>
mutate(
  date = case_when(
    date == "d_lc/d_x_7_1" ~ "2026-07-01",

```

```

      date == "d_lc/d_x_7_16" ~ "2026-07-16",
      date == "d_lc/d_x_7_23" ~ "2026-07-23"
    ),
    gradient = as.numeric(gradient)
  ) |>
left_join(distances_it |> select(site, xmin, xmax),
          by = "site") |>
mutate(site = factor(site, levels = distances_it$site)) |> # FIXED: Uses distances_it inst
filter(!is.na(gradient))

# Gradient plot
plot3_3 <- ggplot(gradient_it_clean) +
  geom_rect(
    aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
    fill = "royalblue4", color = "black", linewidth = 0.3
  ) +
  geom_hline(yintercept = 0, color = "royalblue4", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_it$dist_m,
    labels = distances_it$site
  ) +
  ylim(-4,4) +
  labs(
    title = "Italian Creek (control)",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8)
  )

plot3_3

```

Italian Creek (control)

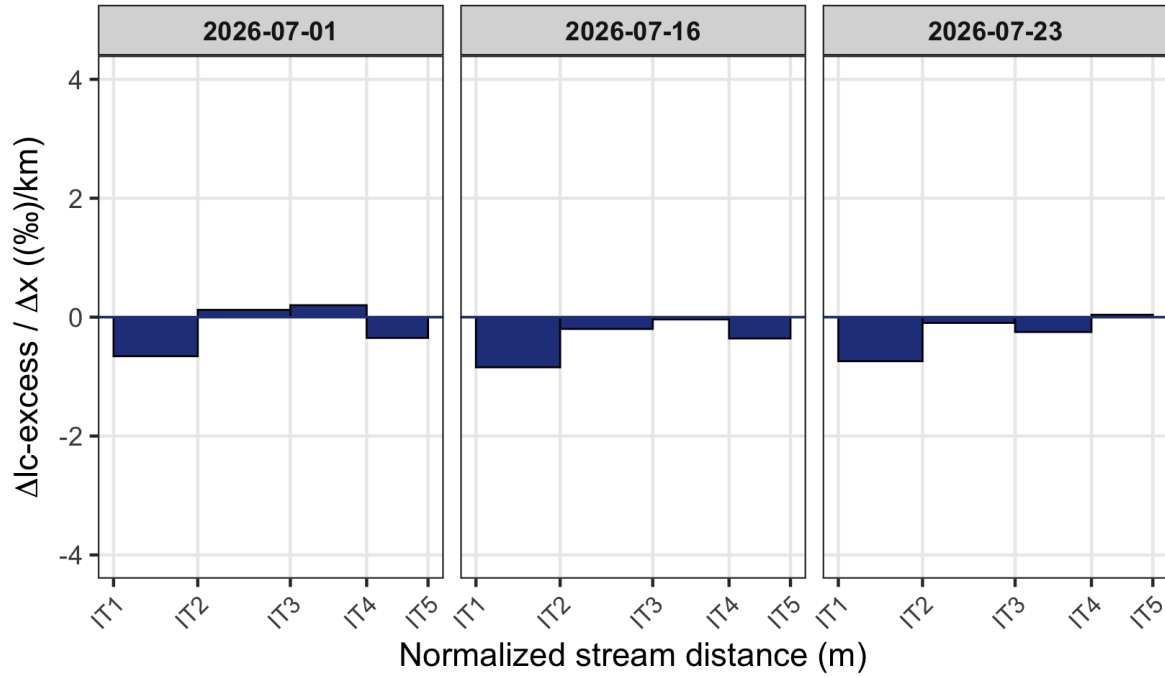


Figure 3

```
figure_3 <- plot3_1 + plot3_2
figure_3 # print the graph
```

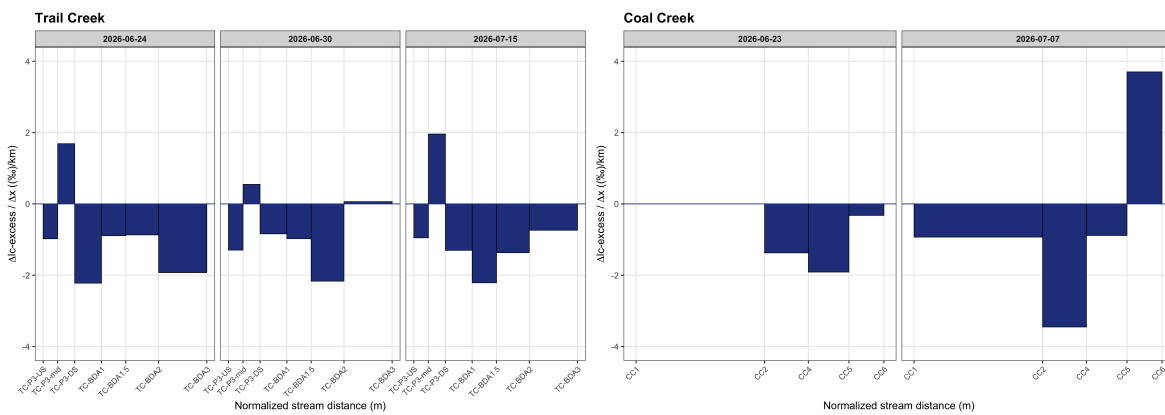
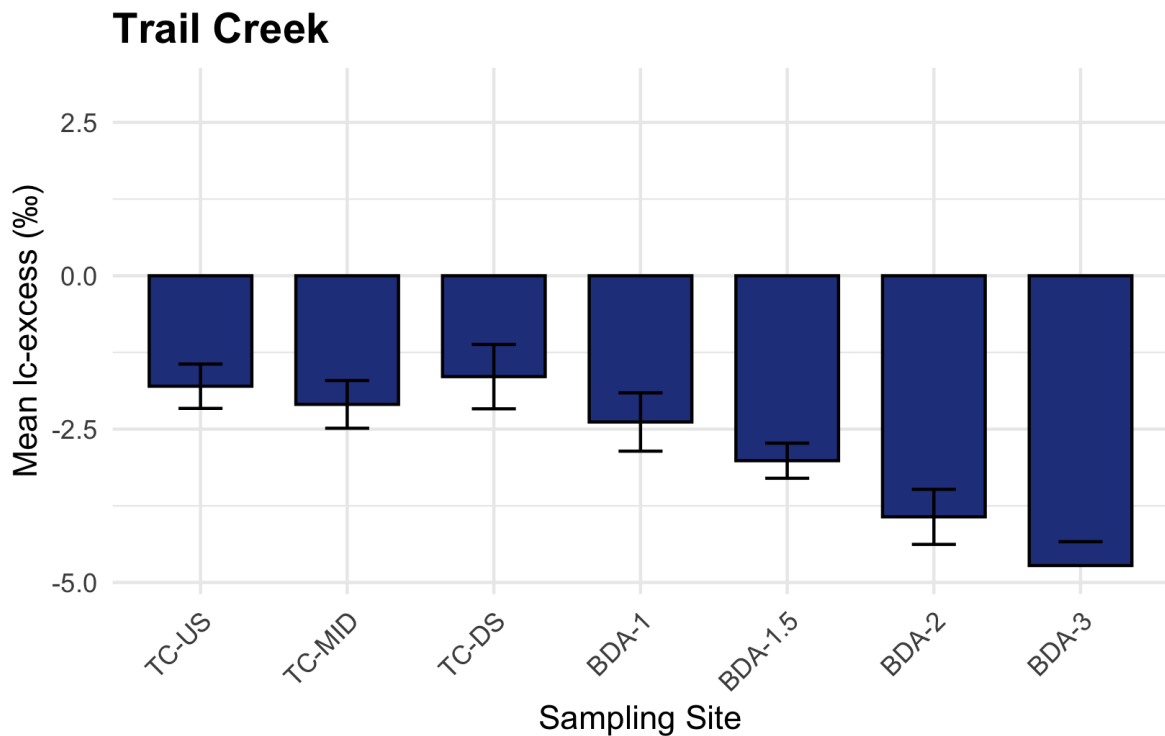


Figure 4

```
# averaging the lc-excess and including standard error
delta_summary <- trail_creek |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (coal creek)
plot4_1 <- ggplot(delta_summary, aes(x = site, y = mean_lc)) +
  geom_col(fill = "royalblue4", color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3,
    color = "black") +
  scale_x_discrete(labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5",
    "tc_2" = "BDA-2", "tc_3" = "BDA-3",
    "tc_ds" = "TC-DS", "tc_mid" = "TC-MID",
    "tc_us" = "TC-US")) +
  ylim(-4.8, 3) +
  theme_minimal() +
  labs(title = "Trail Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (\u2030)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none") # Safely removes the now-empty legend

plot4_1
```



```
# averaging the lc-excess and including standard error
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

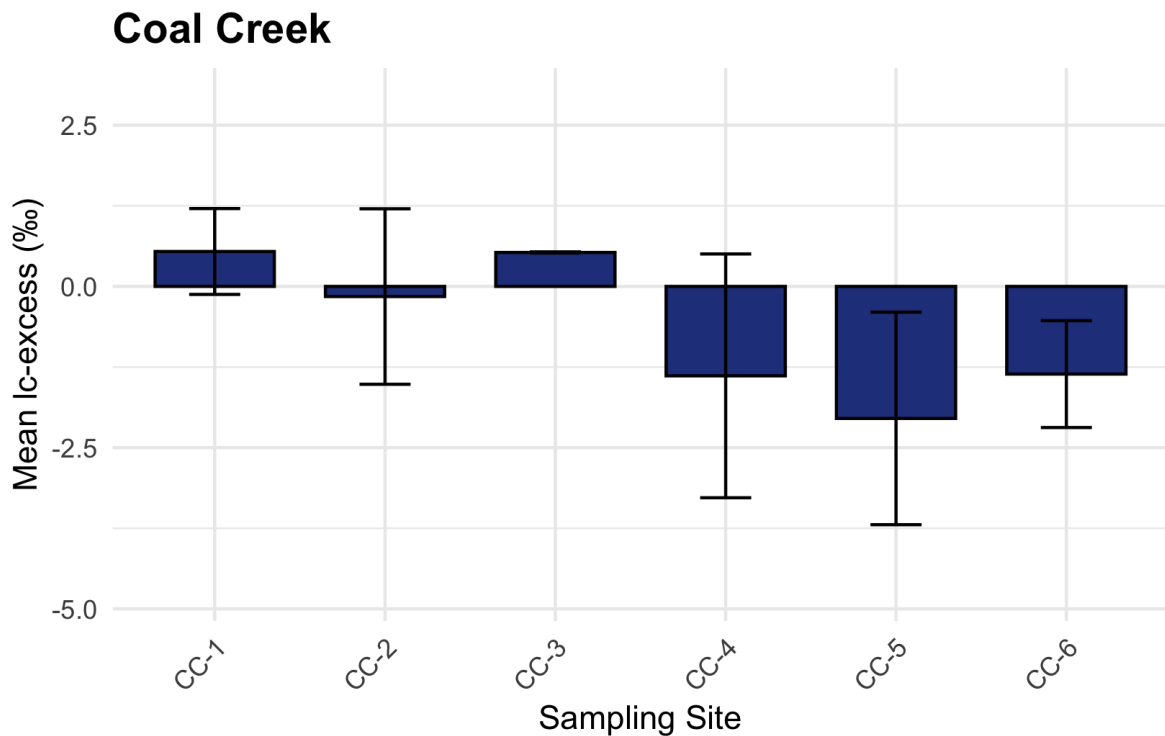
# creating a plot to show the mean lc-excess of each site (coal creek)
plot4_2 <- ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(fill = "royalblue4", color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3, color = "black") +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  ylim(-4.8, 3) +
  theme_minimal() +
  labs(title = "Coal Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (‰)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
```

```

legend.position = "bottom",
legend.key.size = unit(0.4, "cm"),
legend.text = element_text(size = 8),
legend.title = element_text(size = 9),
legend.spacing.x = unit(0.2, "cm"))

```

plot4_2



```

# averaging the lc-excess and including standard error
delta_summary_it <- italian_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (italian creek)
plot4_3 <- ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(fill = "royalblue4", color = "black", width = 0.7) +

```

```

geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
              width = 0.3, color = "black") +
scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3",
                             "italian_4" = "IT-4", "italian_5" = "IT-5")) +
ylim(-4.8, 3) +
theme_minimal() +
labs(title = "Italian Creek (control)",
     x = "Sampling Site",
     y = "Mean lc-excess (\u2030)") +
theme(plot.title = element_text(face = "bold", size = 14),
      axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "bottom",
      legend.key.size = unit(0.4, "cm"),
      legend.text = element_text(size = 8),
      legend.title = element_text(size = 9),
      legend.spacing.x = unit(0.2, "cm"))

```

plot4_3

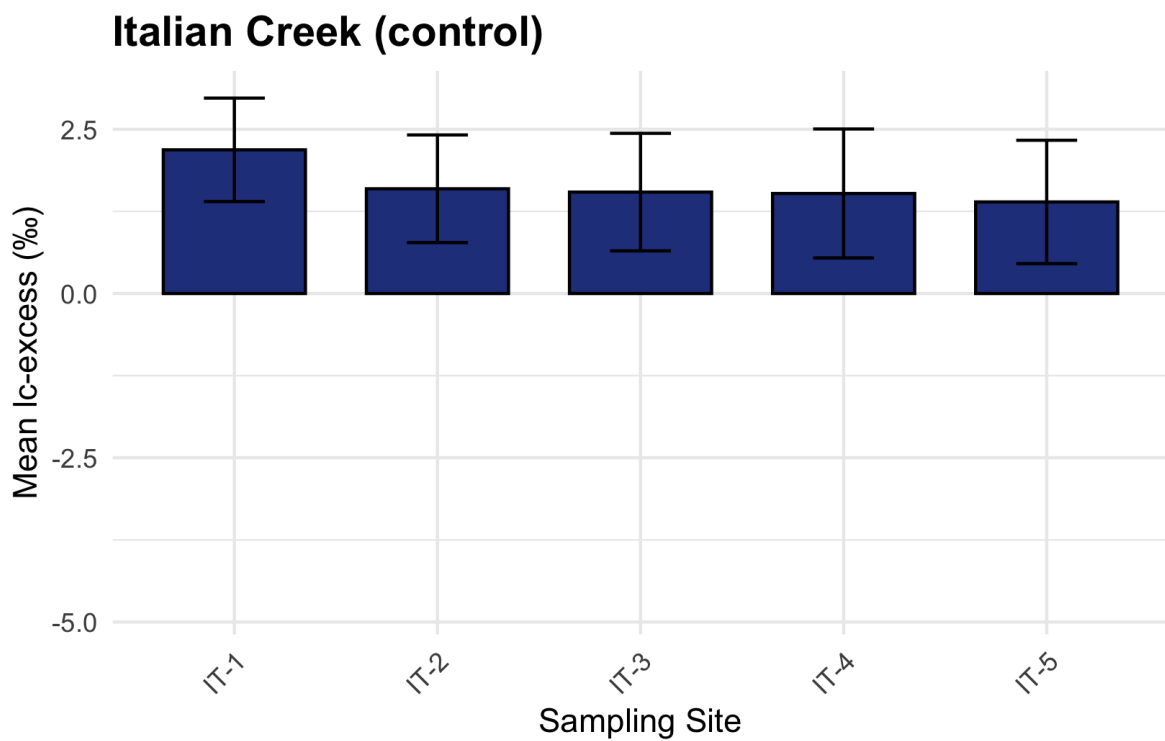


Figure 4:

```
figure_4 <- plot4_3 + plot4_1 + plot4_2

figure_4 # print the graph
```

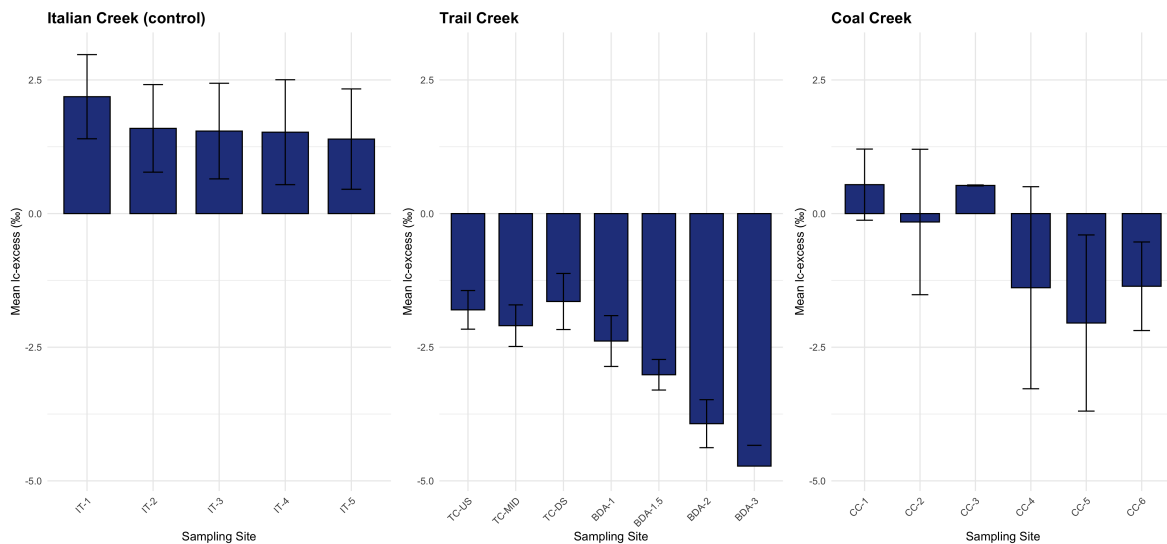
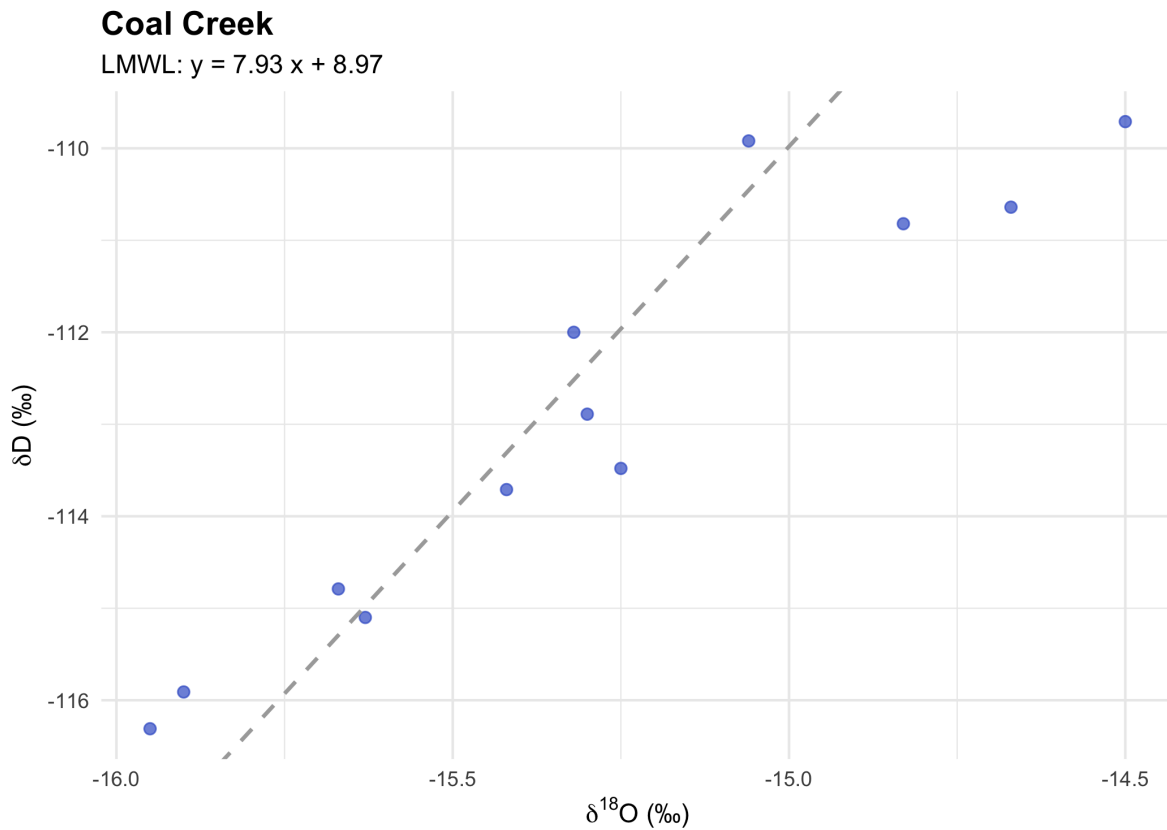


Figure 5:

```
# Coal Creek LMWL calculated from OPIC  $y = 7.93x + 8.97$ 
plot5_1 <- ggplot(coal_creek, aes(x = delta_o, y = delta_d)) +
  geom_point(alpha = 0.7, size = 2, color = "royalblue3") +
  geom_abline(intercept = 8.97, slope = 7.93, linetype = "dashed", color = "darkgray",
              linewidth = 0.8) +
  labs(
    title = "Coal Creek",
    subtitle = "LMWL:  $y = 7.93x + 8.97$ ",
    x = expression(paste(delta^{18}, "O (\u2030)")),
    y = expression(paste(delta, "D (\u2030)"))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_1
```

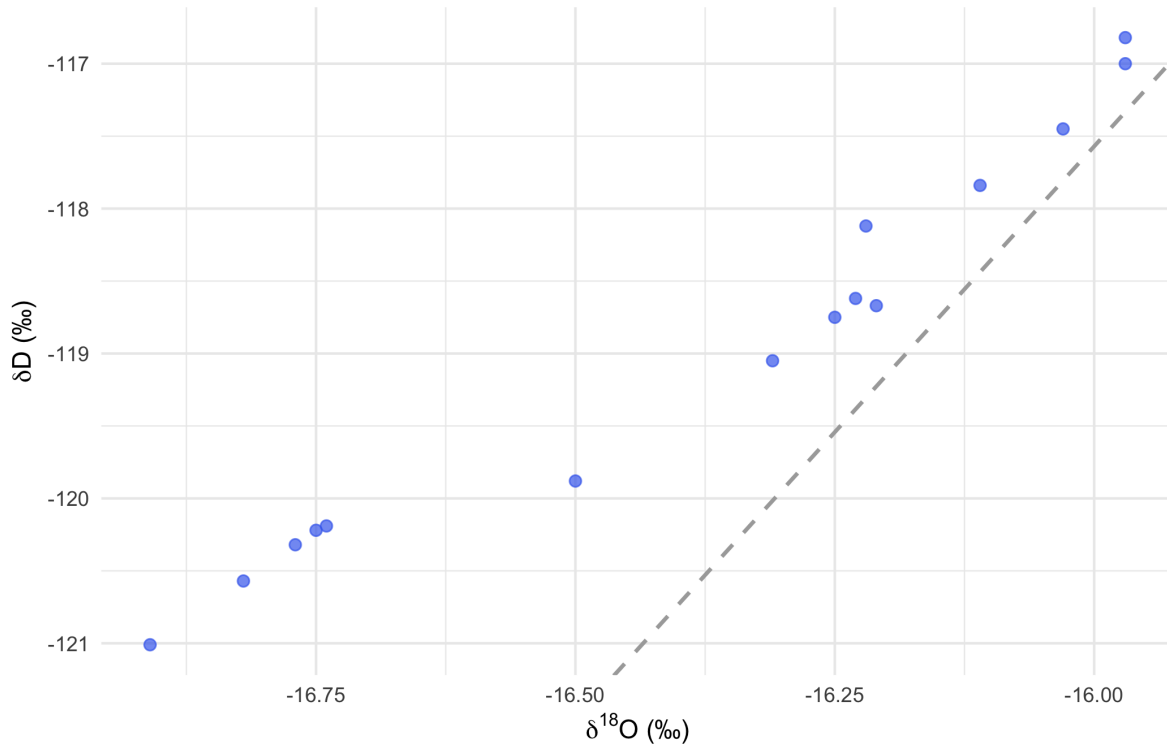


```
# Italian Creek LMWL calculated from OPIC  $y = 7.89x + 8.67$ 

plot5_2 <- ggplot(italian_creek, aes(x = delta_o, y = delta_d)) +
  geom_point(alpha = 0.7, size = 2, color = "royalblue2") +
  geom_abline(intercept = 8.67, slope = 7.89, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Italian Creek",
    subtitle = "LMWL:  $y = 7.89x + 8.67$ ",
    x = expression(paste(delta^{18}, "O (\u2030)")),
    y = expression(paste(delta, "D (\u2030)"))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_2
```

Italian Creek

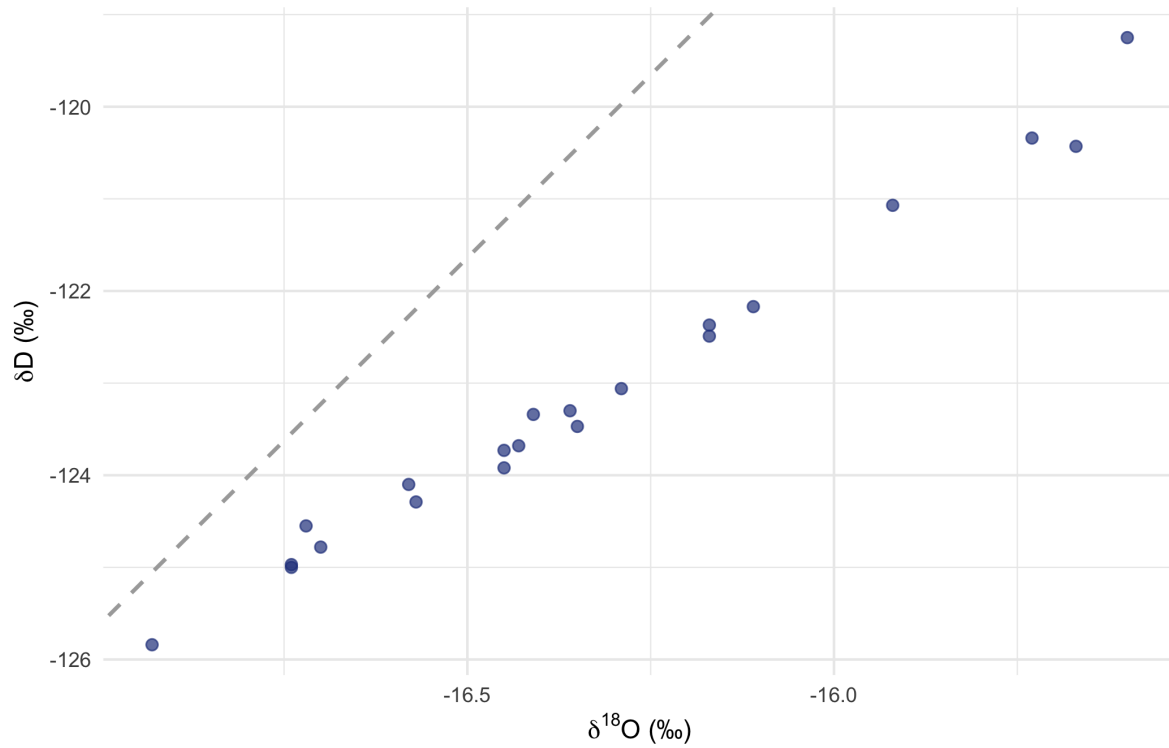
LMWL: $y = 7.89x + 8.67$



```
# Trail Creek River Reach LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
plot5_3 <- ggplot(trail_creek, aes(x = delta_o, y = delta_d)) +
  geom_point(alpha = 0.7, size = 2, color = "royalblue4") +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1)
labs(
  title = "Trail Creek",
  subtitle = "LMWL:  $y = 7.94x + 9.37$ ",
  x = expression(paste(delta^{18}, "O (‰)")),
  y = expression(paste(delta, "D (‰)")),
) +
theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_3
```

Trail Creek

LMWL: $y = 7.94x + 9.37$



```
figure_5.0 <- plot5_3 + plot5_1 + plot5_2  
figure_5.0
```

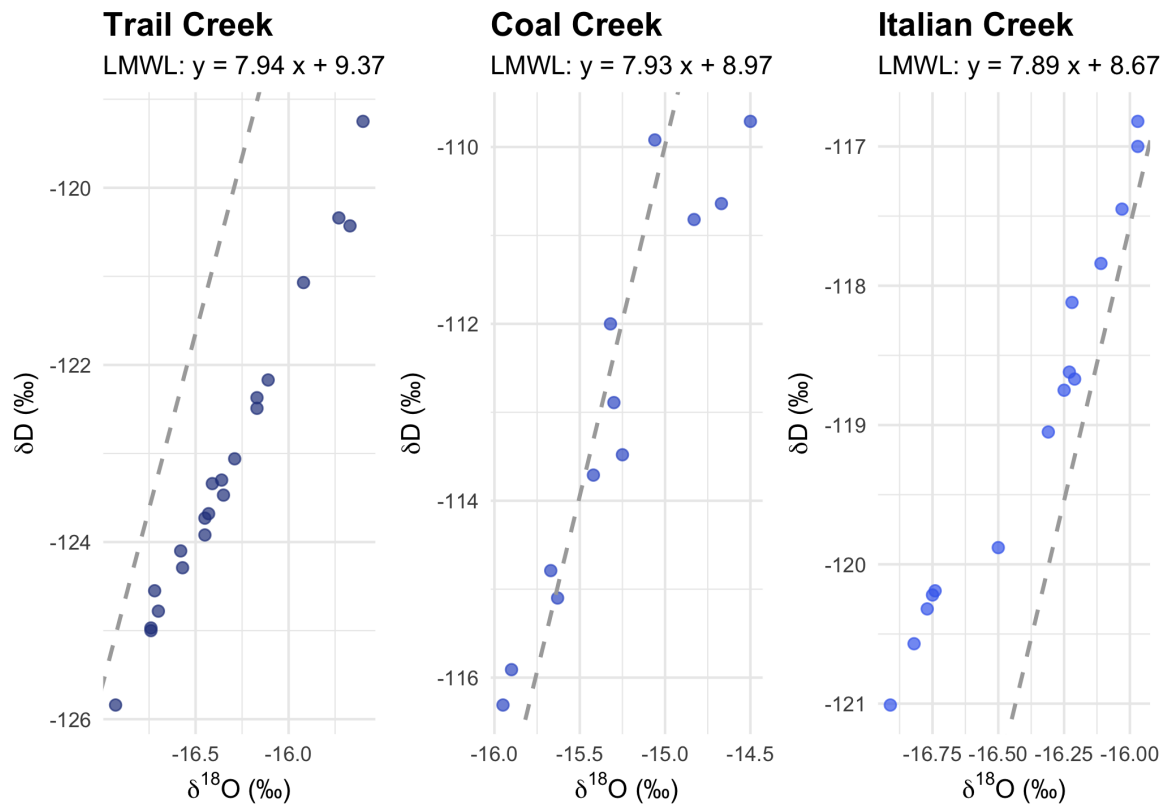
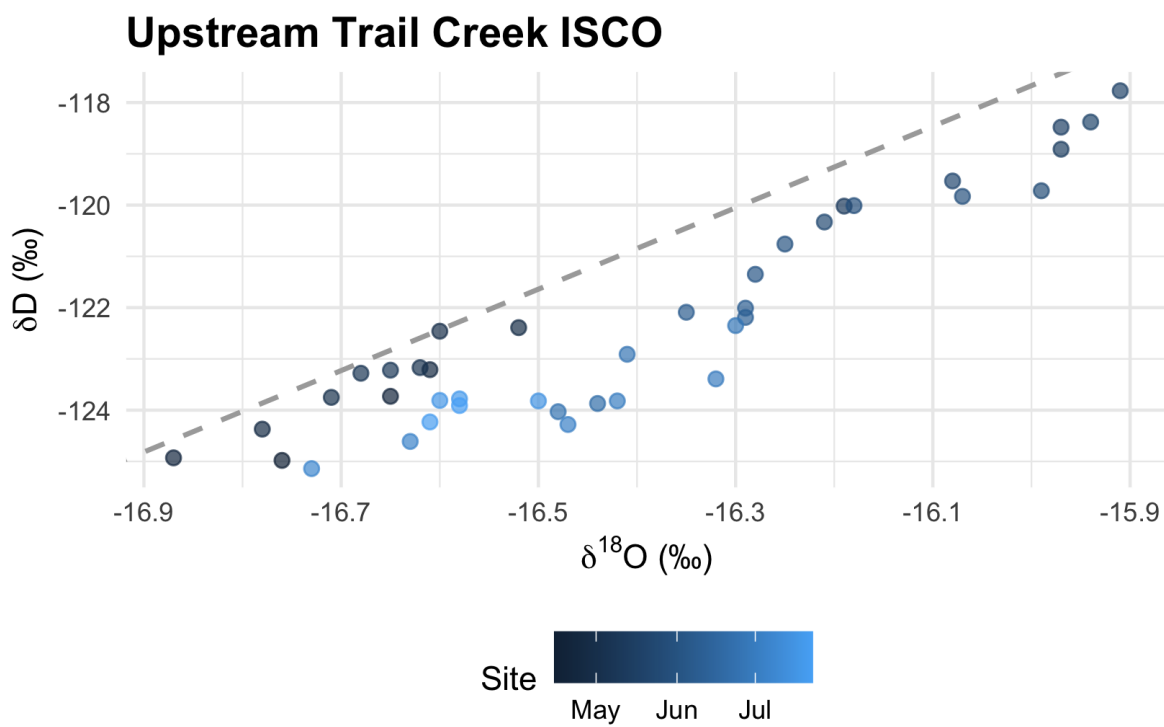


Figure 5.1:

```
# TC ISCO LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
plot5_4 <- ggplot(tc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1)
labs(
  title = "Trail Creek ISCO",
  x = expression(paste(delta^{18}, "O (‰)")),
  y = expression(paste(delta, "D (‰)")),
  color = "Site"
) +
theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "bottom")

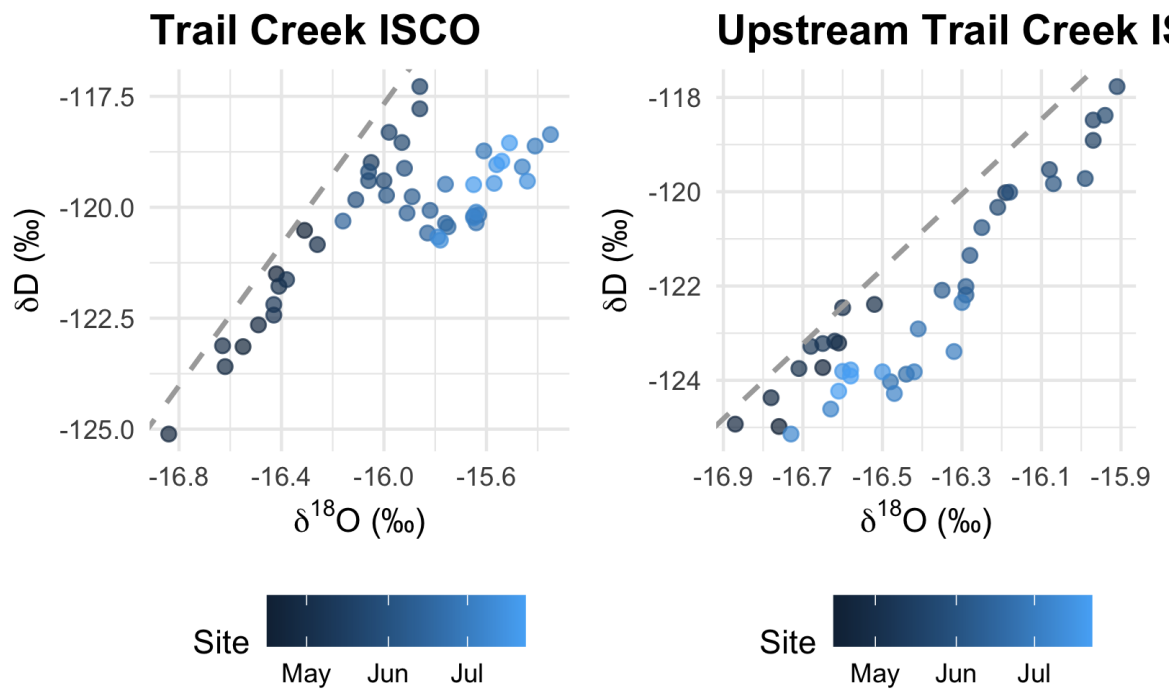
# UTC ISCO LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
```

```
plot5_5 <- ggplot(utc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Upstream Trail Creek ISCO",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "bottom")
plot5_5
```



```
figure_5.1 <- plot5_4 + plot5_5
```

```
figure_5.1
```



TC Discharge

```
model_discharge_tc <- lm(lc_excess ~ discharge, data = discharge_tc_isco)
summary(model_discharge_tc)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.13810	-0.33678	-0.07501	0.29166	1.00884

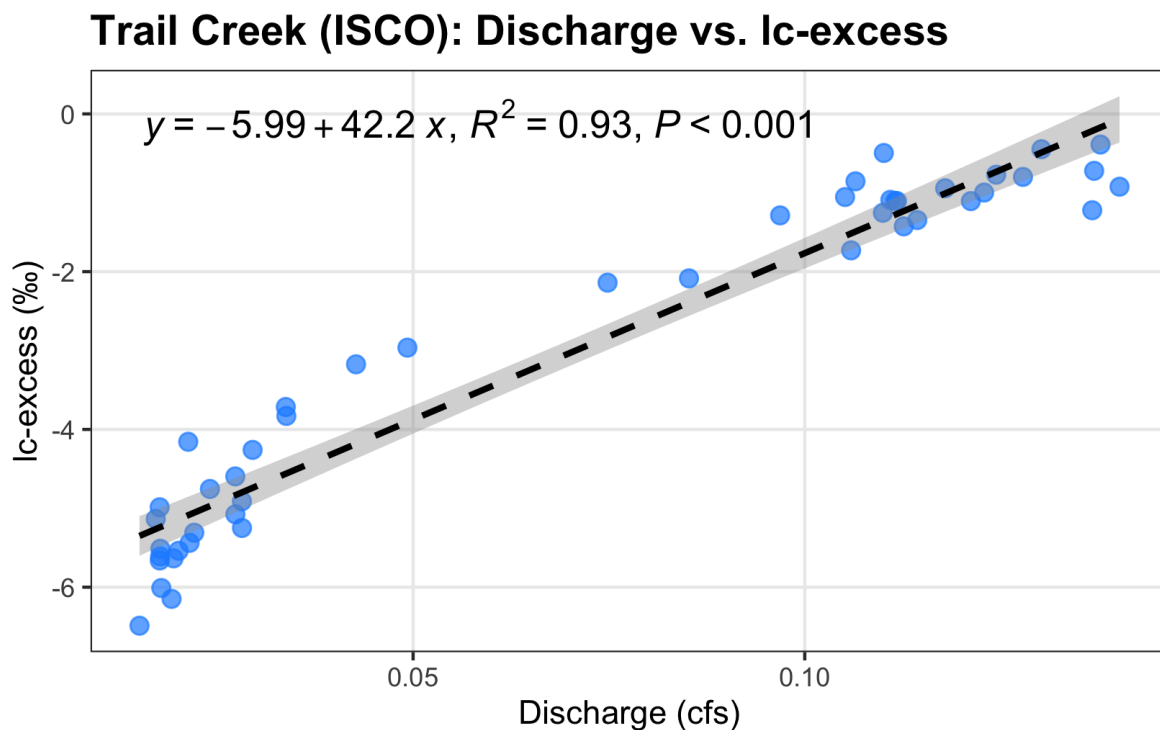
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.986	0.145	-41.28	<2e-16 ***
discharge	42.209	1.719	24.55	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5488 on 44 degrees of freedom
Multiple R-squared: 0.932, Adjusted R-squared: 0.9304
F-statistic: 602.9 on 1 and 44 DF, p-value: < 2.2e-16

```
tc_discharge <- ggplot(discharge_tc_isco, aes(x = discharge, y = lc_excess)) +  
  geom_point(size = 2.5, alpha = 0.7, color = "dodgerblue") +  
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed") +  
  stat_poly_eq(  
    use_label(c("eq", "R2", "P")),  
    formula = y ~ x,  
    label.x = "left", label.y = "top",  
    size = 4.5  
  ) +  
  labs(title = "Trail Creek (ISCO): Discharge vs. lc-excess",  
       x = expression(paste("Discharge (cfs)")),  
       y = "lc-excess (\u2030)") +  
  theme_bw() +  
  theme(plot.title = element_text(face = "bold", size = 14),  
        panel.grid.minor = element_blank())  
  
tc_discharge
```



UTC Discharge

```
model_discharge_utc <- lm(lc_excess ~ discharge, data = discharge_utc_isco)
summary(model_discharge_utc)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_utc_isco)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.39653	-0.58466	0.07884	0.42896	1.38042

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.9751	0.3199	-9.299	2.47e-11 ***
discharge	19.5974	4.1363	4.738	3.00e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.755 on 38 degrees of freedom

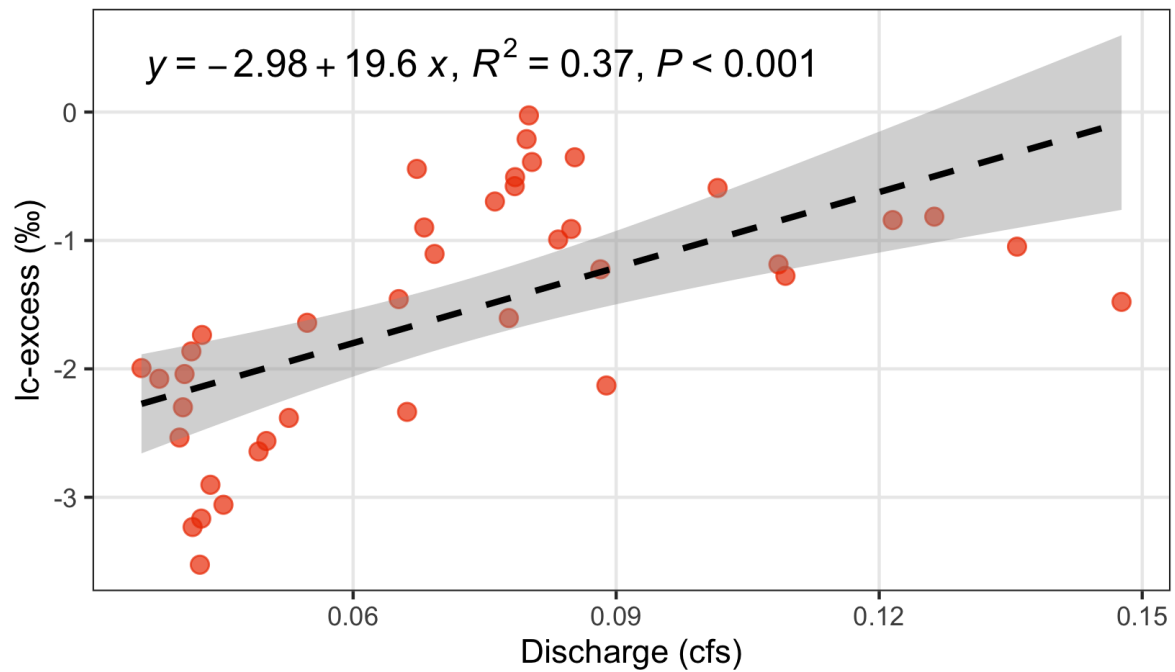
Multiple R-squared: 0.3714, Adjusted R-squared: 0.3548

F-statistic: 22.45 on 1 and 38 DF, p-value: 3e-05

```
utc_discharge <- ggplot(discharge_utc_isco, aes(x = discharge, y = lc_excess)) +
  geom_point(size = 2.5, alpha = 0.7, color = "orangered2") +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed") +
  stat_poly_eq(
    use_label(c("eq", "R2", "P")),
    formula = y ~ x,
    label.x = "left", label.y = "top",
    size = 4.5
  ) +
  labs(title = "Upper Trail Creek (ISCO): Discharge vs. lc-excess",
       x = expression(paste("Discharge (cfs)")),
       y = "lc-excess (\u2030)") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank())

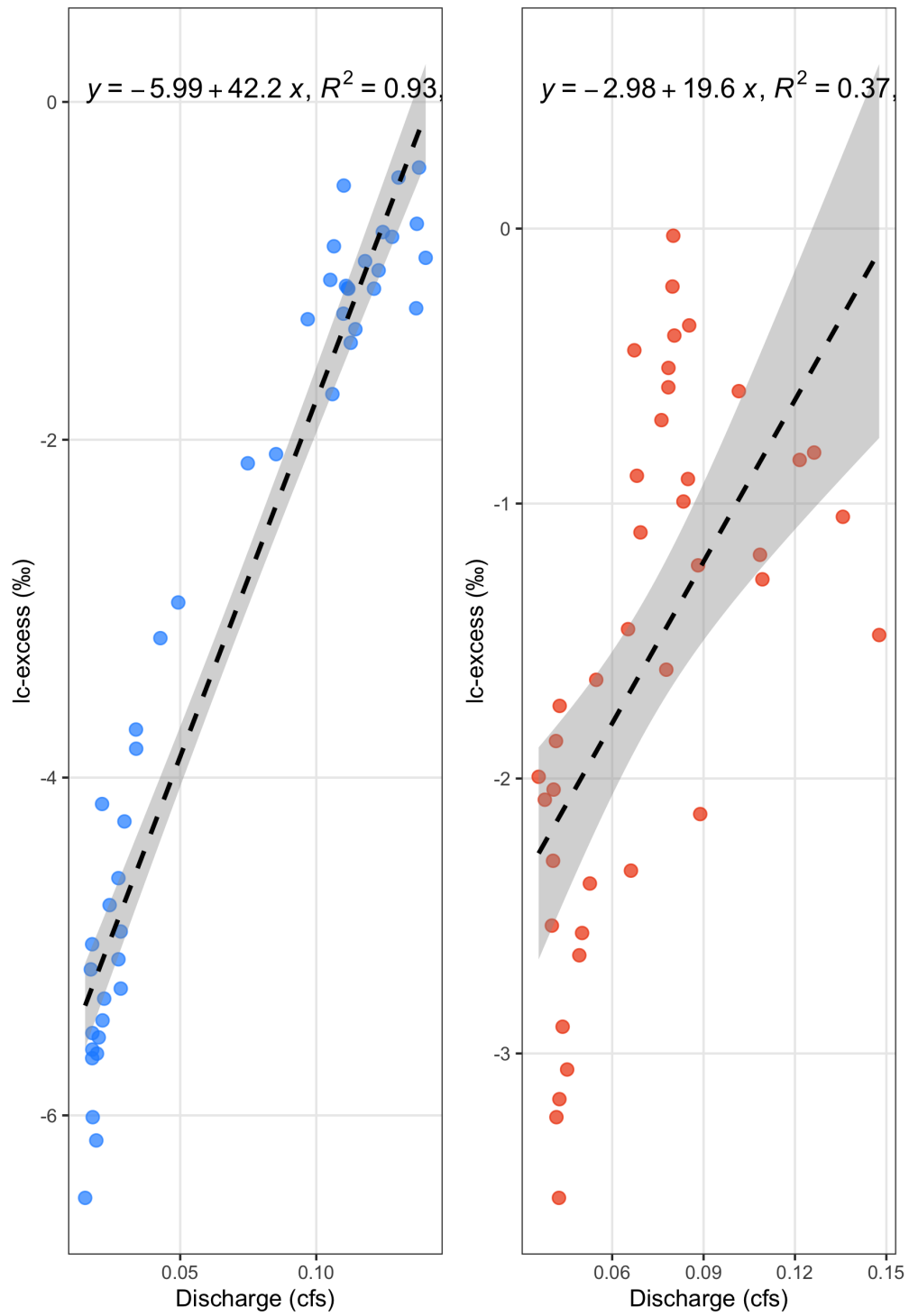
utc_discharge
```

Upper Trail Creek (ISCO): Discharge vs. Ic-excess



```
discharge_figure <- tc_discharge + utc_discharge  
discharge_figure
```

Trail Creek (ISCO): Discharge vs $\delta^{13}\text{C}$ Upper Test Creek (ISCO): Di

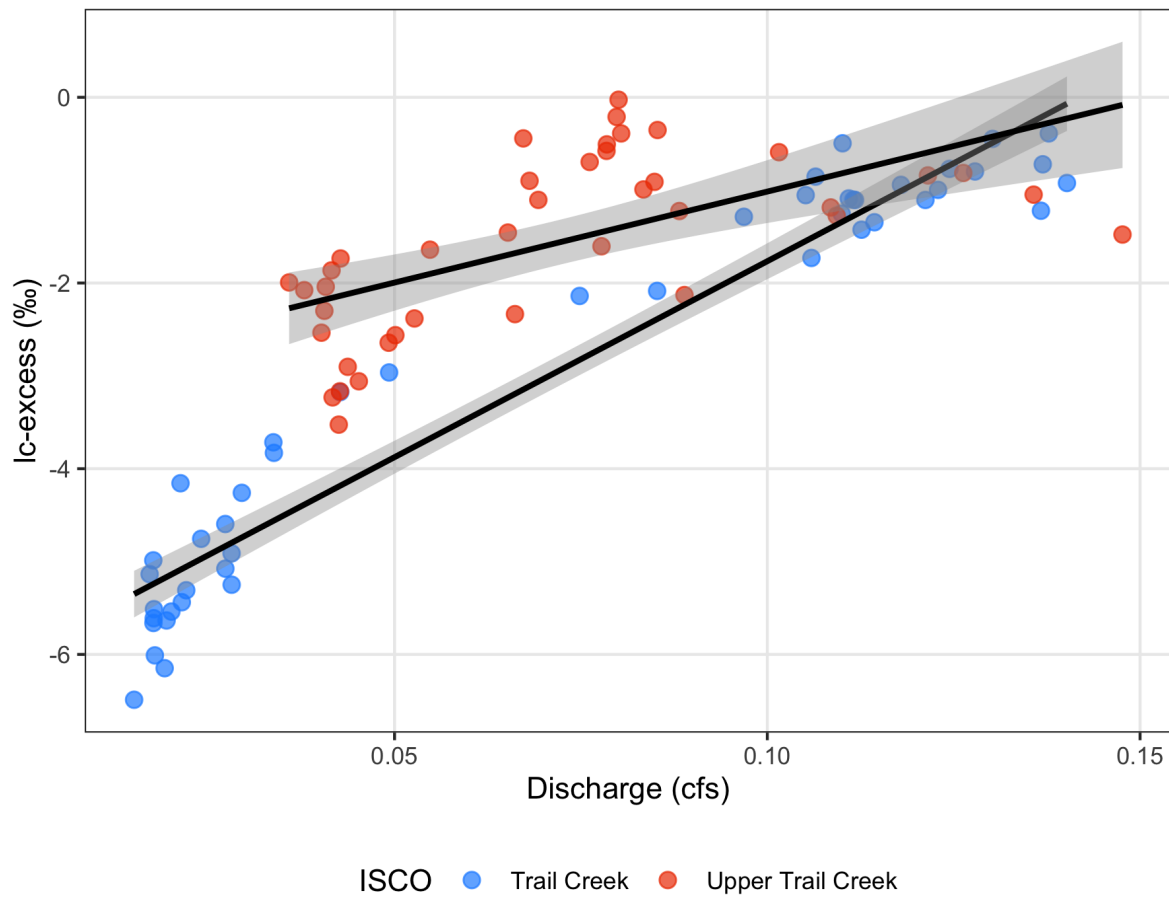


```

ggplot(discharge_isco, aes(x = discharge, y = lc_excess,
                           color = isco,
                           group = isco)) +
geom_point(size = 2.5, alpha = 0.7) +
  scale_color_manual(
    values = c("tc" = "dodgerblue", "utc" = "orangered2"),
    labels = c("tc" = "Trail Creek", "utc" = "Upper Trail Creek")
  ) +
geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black") +
labs(title = "Upper and Lower Trail Creek (ISCO): Discharge vs. lc-excess",
     x = expression(paste("Discharge (cfs)")),
     y = "lc-excess (\u2030)") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "bottom") +
labs(color = "ISCO")

```

Upper and Lower Trail Creek (ISCO): Discharge vs. lc-excess



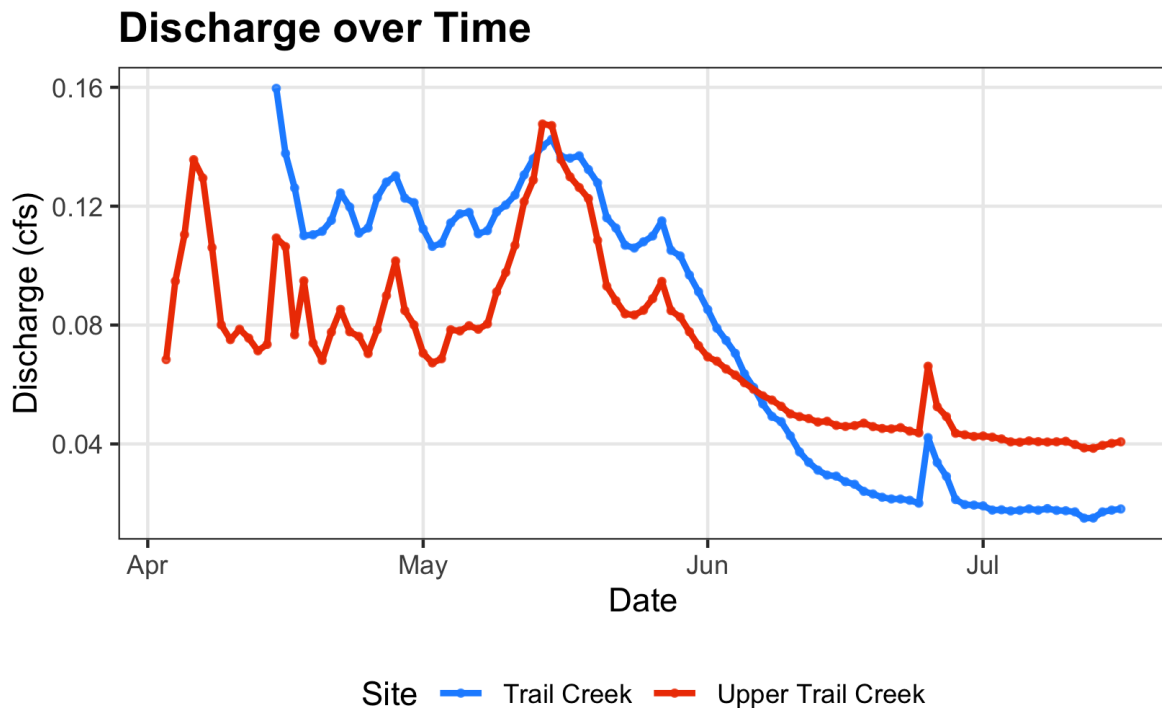
```
# hydrograph dataframe has the discharge, date, and site
# isco_data dataframe has the lc-excess, date, and site
# they have the same information types of date and site

ggplot(data = hydrograph,
  aes(x = date,
    y = discharge,
    color = site,
    group = site)) +
  geom_point(size = 0.8,
    alpha = 0.8) +
  geom_line(linewidth = 1) +
  scale_color_manual(
    values = c("tc" = "dodgerblue", "utc" = "orangered2"),
```

```

  labels = c( "tc" = "Trail Creek", "utc" = "Upper Trail Creek")
) +
labs(title = "Discharge over Time",
     x = "Date",
     y = "Discharge (cfs)",
     color = "Site") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "bottom")

```



```

slope <- 0.01333
intercept <- 0.08

ggplot() +
  # hydrograph
  geom_line(data = hydrograph,
           aes(x = date, y = discharge, color = site, group = site),
           linewidth = 1) +
  # isco_data overlay

```

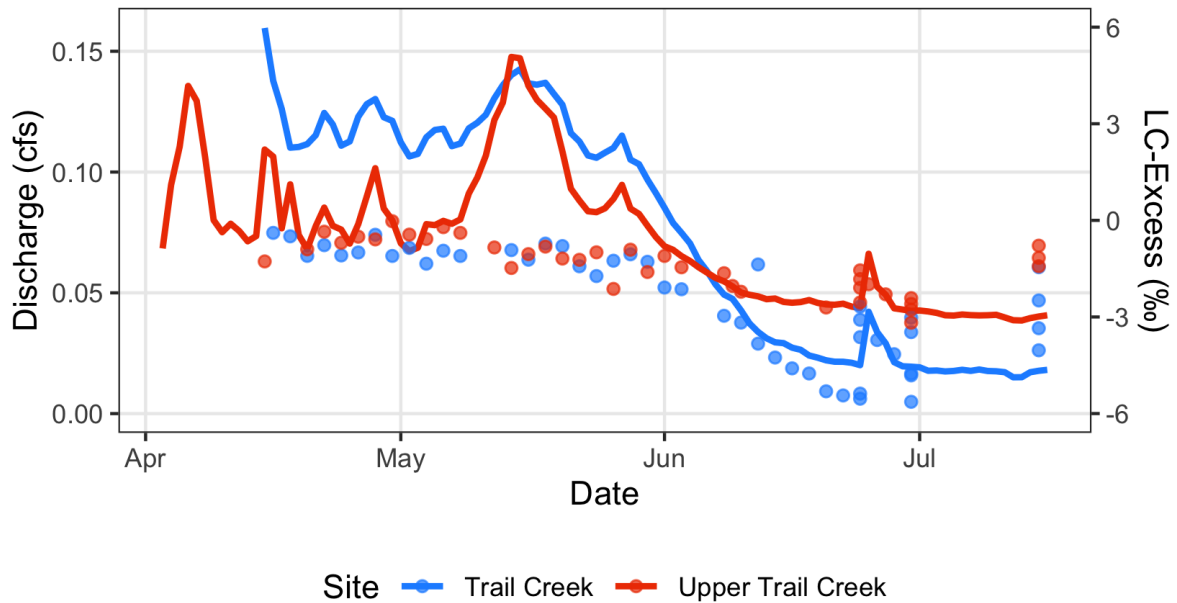
```

geom_point(data = isco_data,
           aes(x = date, y = (lc_excess * slope) + intercept, color = site, group = site),
           shape = 20, # Solid point
           size = 2.5, # Large enough to see clearly
           alpha = 0.7) +
scale_color_manual(
  values = c("tc" = "dodgerblue", "utc" = "orangered2"),
  labels = c("tc" = "Trail Creek", "utc" = "Upper Trail Creek")
) +
scale_y_continuous(
  name = "Discharge (cfs)",
  limits = c(0, 0.16),
  sec.axis = sec_axis(~ (. - intercept) / slope, name = "LC-Excess (\u2030)")
) +
labs(title = "Discharge and LC-Excess over time",
     subtitle = "Lines show discharge; points show lc-excess samples",
     x = "Date",
     color = "Site") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "bottom") +
# cleans up the legend box so it displays both a line and a triangle together
guides(color = guide_legend(override.aes = list(shape = c(20, 20), linetype = c(1, 1))))

```

Discharge and LC-Excess over time

Lines show discharge; points show lc-excess samples

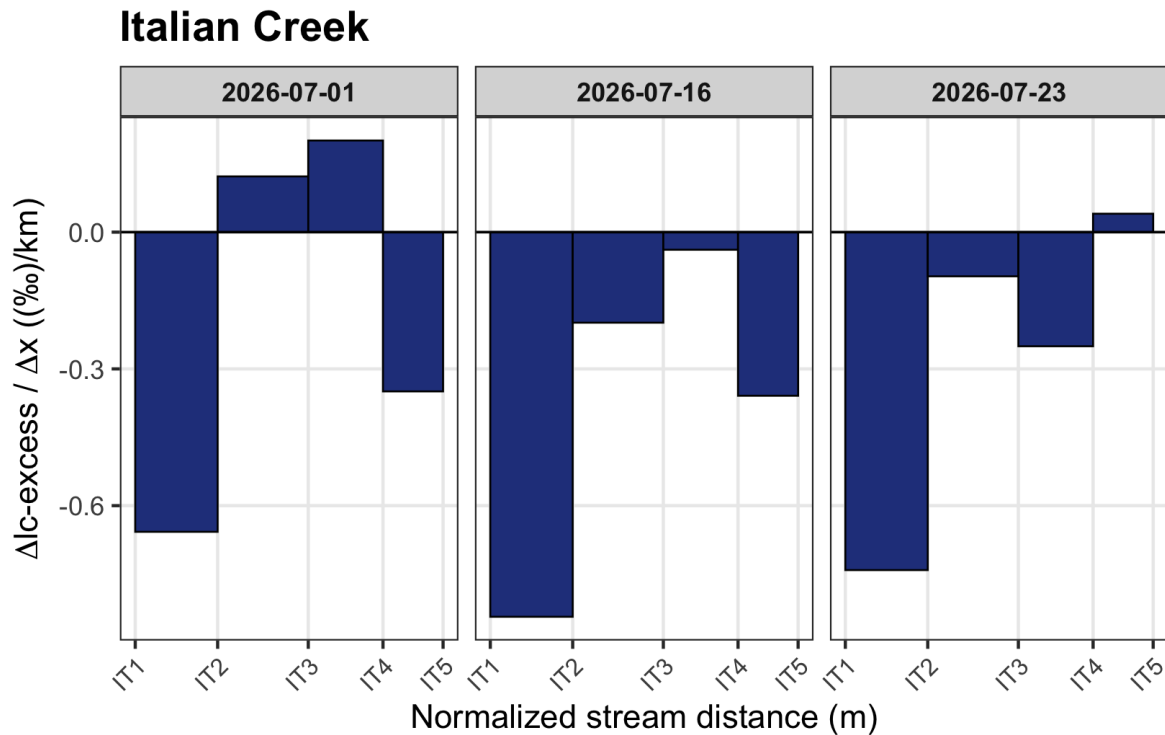


```
# gradient plot for it
plot6_1 <- ggplot(gradient_it_clean) +
  geom_rect(
    aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
    fill = "royalblue4", color = "black", linewidth = 0.3
  ) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_it$dist_m,
    labels = distances_it$site
  ) +
  labs(
    title = "Italian Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
  )
```

```

strip.text = element_text(face = "bold"),
axis.text.x = element_text(angle = 45, hjust = 1, size = 8)
)
plot6_1

```



```

gradient_cum_tc <- gradient_tc |>
  select(site, cumulative_dams,
         `Δlc-excess/Δcum_dam_length_6-24`, `Δlc-excess/Δcum_dam_length_6-30`, `Δlc-excess/Δcum_dam_length_7-15`)
  pivot_longer(
    cols = c(`Δlc-excess/Δcum_dam_length_6-24`, `Δlc-excess/Δcum_dam_length_6-30`, `Δlc-excess/Δcum_dam_length_7-15`),
    names_to = "date",
    values_to = "gradient"
  ) |>
  mutate(
    date = case_when(
      date == "Δlc-excess/Δcum_dam_length_6-24" ~ "2026-06-24",
      date == "Δlc-excess/Δcum_dam_length_6-30" ~ "2026-06-30",
      date == "Δlc-excess/Δcum_dam_length_7-15" ~ "2026-07-15"
    ),
    gradient = as.numeric(gradient),
  )

```

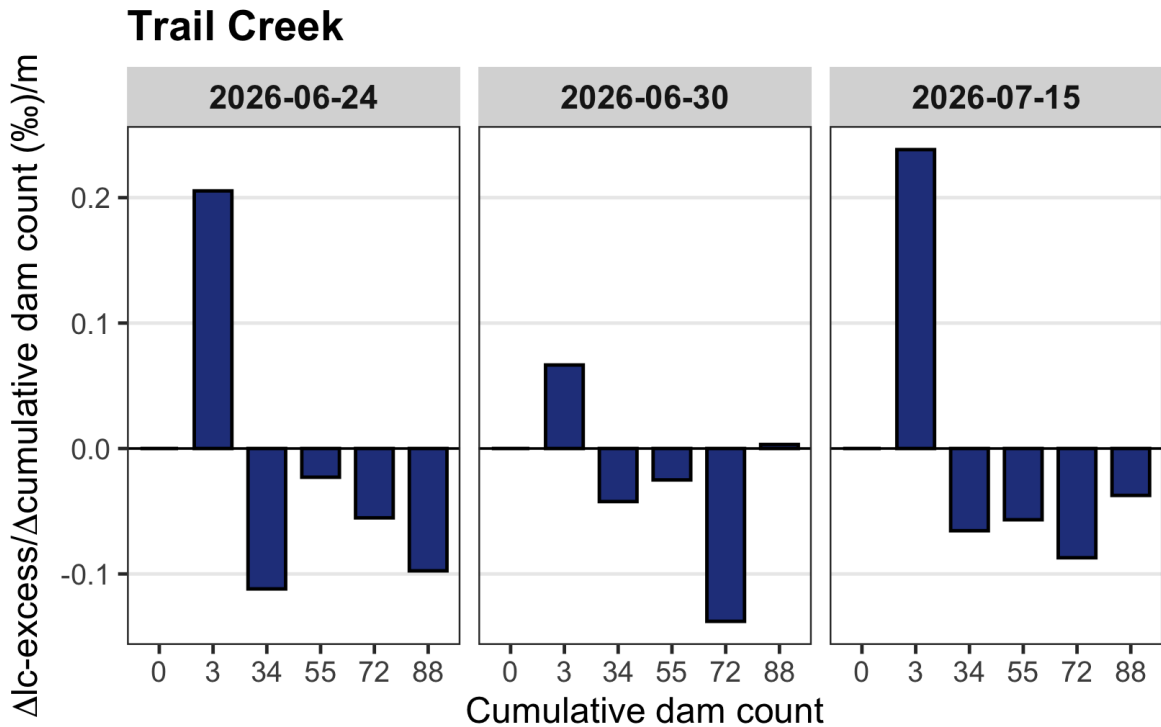
```

    cumulative_dams = as.numeric(cumulative_dams),
    site = factor(site, levels = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS",
                                   "TC-BDA1", "TC-BDA1.5", "TC-BDA2", "TC-BDA3"))
  ) |>
  filter(!is.na(gradient))

plot6_2 <- ggplot(gradient_cum_tc, aes(x = site, y = gradient)) + # Removed fill = site
  geom_col(fill = "royalblue4", color = "black", width = 0.7) + # Added fill = "royalblue4"
  geom_hline(yintercept = 0, color = "black", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_tc$cumulative_dams, gradient_cum_tc$site)
    lookup[x]
  }) +
  labs(title = "Trail Creek",
       x = "Cumulative dam count",
       y = expression(Delta * "lc-excess/" * Delta * "cumulative dam count (" * "\u2030" * ")/m
  ) +
  theme_bw(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        strip.text = element_text(face = "bold", size = 11),
        strip.background = element_rect(fill = "gray85", color = NA),
        panel.grid.minor = element_blank(),
        panel.grid.major.x = element_blank(),
        legend.position = "none" # Removes the now empty legend box
  )

plot6_2

```



```

gradient_cum_cc <- gradient_cc |>
  select(site, cumulative_dams, `Δlc-excess/Δavg_dam_length_6-23 (ppb/m)`,
    `Δlc-excess/Δcum_dam_length_7-7 (ppb/m)`) |>
  pivot_longer(
    cols = c(`Δlc-excess/Δavg_dam_length_6-23 (ppb/m)`, `Δlc-excess/Δcum_dam_length_7-7 (ppb/m)`),
    names_to = "date", values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δavg_dam_length_6-23 (ppb/m)" ~ "2026-06-23",
    date == "Δlc-excess/Δcum_dam_length_7-7 (ppb/m)" ~ "2026-07-07"),
    gradient = as.numeric(gradient), cumulative_dams = as.numeric(cumulative_dams),
    site = factor(site)) |>
  filter(!is.na(gradient))

plot6_3 <- ggplot(gradient_cum_cc, aes(x = site, y = gradient)) + # Removed fill = site
  geom_col(fill = "royalblue4", color = "black", width = 0.7) + # Added fill = "royalblue4"
  geom_hline(yintercept = 0, color = "black", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_cc$cumulative_dams, gradient_cum_cc$site)
    lookup[x]
  })

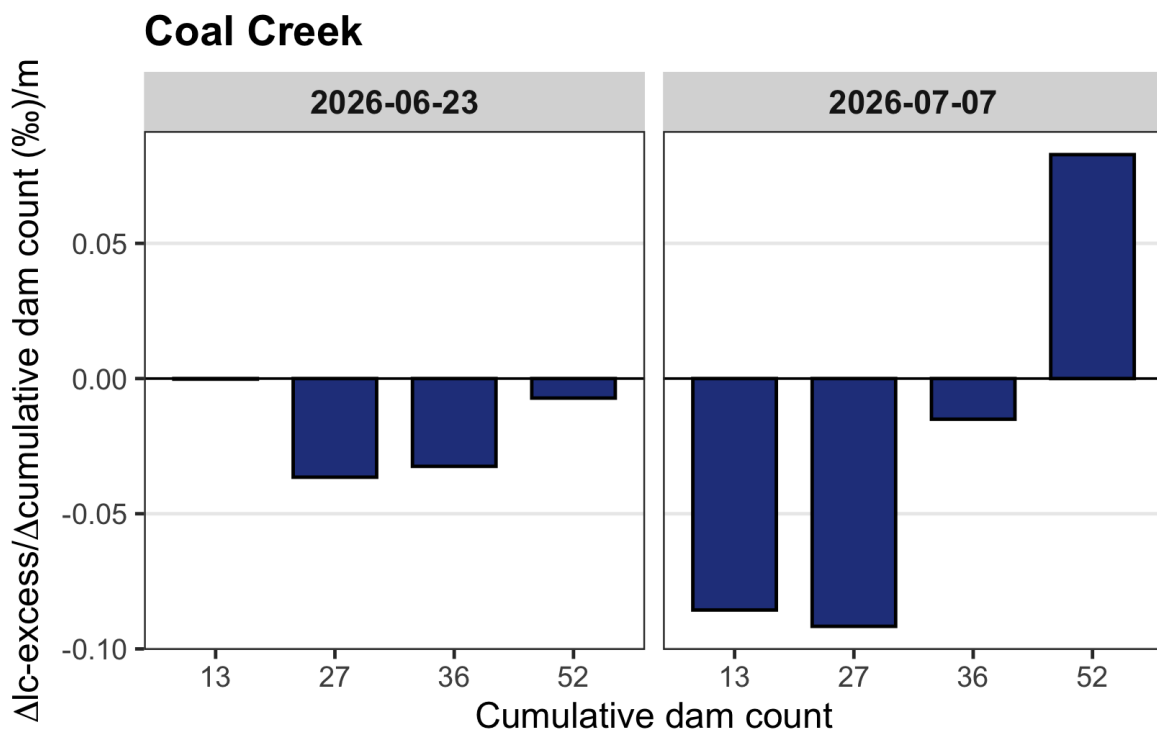
```

```

}) +
labs(title = "Coal Creek",
     x = "Cumulative dam count",
     y = expression(Delta * "lc-excess/" * Delta * "cumulative dam count (" * "\u2030" * ")/m
) +
theme_bw(base_size = 12) +
theme(plot.title = element_text(face = "bold", size = 14),
      strip.text = element_text(face = "bold", size = 11),
      strip.background = element_rect(fill = "gray85", color = NA),
      panel.grid.minor = element_blank(),
      panel.grid.major.x = element_blank(),
      legend.position = "none" # Removes the now empty legend box
)

```

plot6_3



```
figure_6 <- plot6_2 + plot6_3
```

figure_6

